

BIOL-1 Practice Test 03

Exam 02 Preparation (Modules 7–11)

Instructions: This practice test covers material from Modules 7 through 11. Answer all questions to the best of your ability to prepare for Exam 02.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

Module 7: Molecular Genetics

1. The enzyme that unwinds the DNA double helix during replication is: A) DNA polymerase B) RNA polymerase C) Helicase D) Ligase
2. Okazaki fragments are synthesized on the: A) Leading strand B) Lagging strand C) mRNA transcript D) tRNA molecule
3. Transcription is the process of converting: A) RNA into a protein B) DNA into RNA C) DNA into a new DNA molecule D) Protein into an observable trait
4. Which part of a primary mRNA transcript is removed during RNA splicing before it leaves the nucleus? A) Exons B) Introns C) Codons D) Promoters
5. A mutation that shifts the entire reading frame of the ribosome is called a: A) Missense mutation B) Point mutation C) Frameshift mutation D) Silent mutation

Module 8: Cellular Genetics (Mitosis & Meiosis)

6. Which phase of the cell cycle is characterized by DNA replication? A) G1 phase B) S phase C) G2 phase D) M phase

7. After a single human cell undergoes meiosis, the resulting cells are: A) 2 diploid, genetically identical cells B) 2 haploid, genetically identical cells C) 4 diploid, genetically distinct cells D) 4 haploid, genetically distinct cells
8. Crossing over, which increases genetic diversity, occurs during: A) Prophase of Mitosis B) Prophase I of Meiosis C) Metaphase II of Meiosis D) Anaphase I of Meiosis
9. Nondisjunction during meiosis can lead to conditions such as Down syndrome, which is caused by trisomy of chromosome: A) 18 B) 21 C) 23 D) Y
10. The p53 protein is heavily involved in: A) Attaching spindle fibers to kinetochores B) Stopping the cell cycle if DNA is damaged C) Breaking down the nuclear envelope D) Forming the cleavage furrow

Module 9: Inheritance Genetics

11. The physical appearance or observable trait of an organism is its: A) Genotype B) Phenotype C) Allele D) Karyotype
12. In a monohybrid cross between two heterozygous parents ($Aa \times Aa$), what is the expected phenotypic ratio for a completely dominant trait? A) 1:1 B) 1:2:1 C) 3:1 D) 9:3:3:1
13. A true-breeding red snapdragon is crossed with a true-breeding white snapdragon, producing offspring that are all pink. This is an example of: A) Codominance B) Incomplete dominance C) Polygenic inheritance D) Sex-linked inheritance
14. Human blood types exhibit which type(s) of inheritance pattern(s)? A) Simple dominance only B) Codominance and multiple alleles C) Incomplete dominance D) Sex-linked recessive
15. Why are sex-linked recessive traits more common in human males than females? A) Males have a stronger Y chromosome. B) Males only have one X chromosome, so a single recessive allele determines the phenotype. C) Females lack the genes required for these traits entirely. D) The X chromosome in males is larger than in females.

Module 10: Epigenetics

- 16.** Epigenetics refers to changes in gene expression that: A) Alter the underlying DNA sequence permanently. B) Do not change the DNA sequence but can be passed to daughter cells. C) Are only caused by environmental toxins. D) Only occur in prokaryotes.
- 17.** DNA methylation generally has what effect on gene expression? A) It increases transcription. B) It decreases or silences transcription. C) It has no effect on transcription. D) It causes the DNA to mutate rapidly.
- 18.** The lac operon in *E. coli* allows the bacteria to produce enzymes to digest lactose only when lactose is present. This is an example of: A) Epigenetic inheritance B) Gene regulation C) A frameshift mutation D) A sex-linked trait
- 19.** Acetylation of histones usually results in: A) Tightly packed chromatin and decreased transcription. B) Loosely packed chromatin and increased transcription. C) Immediate degradation of the DNA. D) The splicing of introns out of mRNA.
- 20.** The random inactivation of one X chromosome in female mammalian cells results in physical traits like tortoiseshell coloring in cats. The inactivated X chromosome condenses into a structure known as a: A) Centrosome B) Telomere C) Barr body D) Nucleosome

Module 11: Genomics & Biotechnology

- 21.** The primary function of restriction enzymes in biotechnology is to: A) Amplify DNA sequences. B) Cut DNA at highly specific sequence sites. C) Analyze protein folding. D) Glue Okazaki fragments together.
- 22.** Polymerase Chain Reaction (PCR) is used to: A) Separate DNA fragments by size. B) Rapidly make millions of copies of a specific DNA segment. C) Insert foreign DNA into a bacterial plasmid. D) Translate mRNA into a polypeptide chain in a test tube.
- 23.** During gel electrophoresis, DNA fragments move toward the positive electrode because: A) DNA is positively charged. B) DNA has a neutral charge but is pulled by gravity. C) the phosphate backbone gives DNA a negative charge. D) agarose gel attracts DNA chemically.
- 24.** The CRISPR-Cas9 system was originally discovered in: A) Viruses as an attack mechanism against hosts. B) Bacteria as a rudimentary cellular immune system against viral

bacteriophages. C) Human liver cells to repair damaged DNA. D) Plants to resist fungal infections.

25. Transgenic organisms, such as bacteria engineered to produce human insulin, are possible because: A) Bacteria and humans have identical DNA sequences. B) All organisms share a nearly universal genetic code. C) Human genes naturally mutate into bacterial genes. D) Transcription only works in eukaryotic laboratory cells.

Part B: Fill in the Blank (10 questions)

26. In RNA, the nitrogenous base *_ replaces the Thymine found in DNA.* 27. *According to the Central Dogma of molecular biology, DNA is transcribed into _*, which is then translated into a protein. 28. The division of the cytoplasm at the very end of mitosis or meiosis is called *_*. 29. *The presence of three copies of a chromosome instead of the normal two is referred to as a _*. 30. A gene variant that causes death when inherited in homozygous form is known as a *_ allele*. 31. *An individual with two different alleles for a specific trait (e.g., Bb) is called _*. 32. In gene regulation, a(n) *_ is a cluster of genes under the control of a single promoter, commonly found in prokaryotes.* 33. *Identical twins have the exact same DNA sequence, but as they age, differences in their gene expression accumulate due to _ changes.* 34. In gel electrophoresis, *_ fragments of DNA travel through the gel matrix faster and farther than larger fragments.* 35. *The use of an organism, or a component of an organism, to make a product or process for human use is known as _*.

Part C: Short Answer (4 questions)

36. (Module 7) Write the complementary DNA sequence AND the transcribed mRNA sequence for the following DNA template strand: 3'-TAC GGG TTA ACG ACT-5'.

37. (Module 8 & 9) Explain the difference between Mitosis and Meiosis in terms of their overall purpose for a multicellular organism like a human. Briefly link how Meiosis connects to Mendel's Law of Segregation.

38. (Module 10) Describe how it is possible for a muscle cell and a nerve cell in your body to look and function so differently, despite containing the exact same DNA.

39. (Module 11) Choose ONE biotechnology tool discussed in class (e.g., PCR, Gel Electrophoresis, CRISPR, Recombinant Plasmids). Explain what it does and provide one real-world application or example of its use.

End of Practice Test 03

Answer Key

Part A: Multiple Choice

1. C (Helicase)
2. B (Lagging strand)
3. B (DNA into RNA)
4. B (Introns)
5. C (Frameshift mutation)
6. B (S phase)
7. D (4 haploid, genetically distinct cells)
8. B (Prophase I of Meiosis)
9. B (21)
10. B (Stopping the cell cycle if DNA is damaged)
11. B (Phenotype)
12. C (3:1)
13. B (Incomplete dominance)
14. B (Codominance and multiple alleles)
15. B (Males only have one X chromosome)
16. B (Do not change the DNA sequence but can be passed to daughter cells)
17. B (It decreases or silences transcription)
18. B (Gene regulation)
19. B (Loosely packed chromatin and increased transcription)
20. C (Barr body)
21. B (Cut DNA at highly specific sequence sites)
22. B (Rapidly make millions of copies of a specific DNA segment)
23. C (the phosphate backbone gives DNA a negative charge)
24. B (Bacteria as a rudimentary cellular immune system...)
25. B (All organisms share a nearly universal genetic code)

Part B: Fill in the Blank

1. Uracil

2. mRNA (or RNA)
3. Cytokinesis
4. Trisomy
5. Lethal
6. Heterozygous
7. Operon
8. Epigenetic
9. Smaller (or shorter)
10. Biotechnology

Part C: Short Answer (Key Concepts)

1. **Complementary DNA:** 5'-ATG CCC AAT TGC TGA-3'
mRNA: 5'-AUG CCC AAU UGC UGA-3'
2. **Mitosis:** cellular division for growth and repair, produces identical diploid somatic cells.
Meiosis: division for reproduction, produces genetically unique haploid gametes.
Mendel's Law of Segregation occurs during Meiosis I when homologous chromosomes separate.
3. **Gene Regulation / Epigenetics:** Both cells have identical DNA, but specific genes are turned on or off based on the cell type. Certain regions of chromatin might be tightly packed (methylated) in a nerve cell but loosely packed (acetylated) in a muscle cell.
4. Example answers include: **PCR** (amplifies DNA for forensics/testing), **CRISPR** (edits specific DNA sequences for gene therapy), **Gel Electrophoresis** (separates DNA by size for DNA fingerprinting), **Recombinant Plasmids** (inserting human insulin gene into bacteria).